

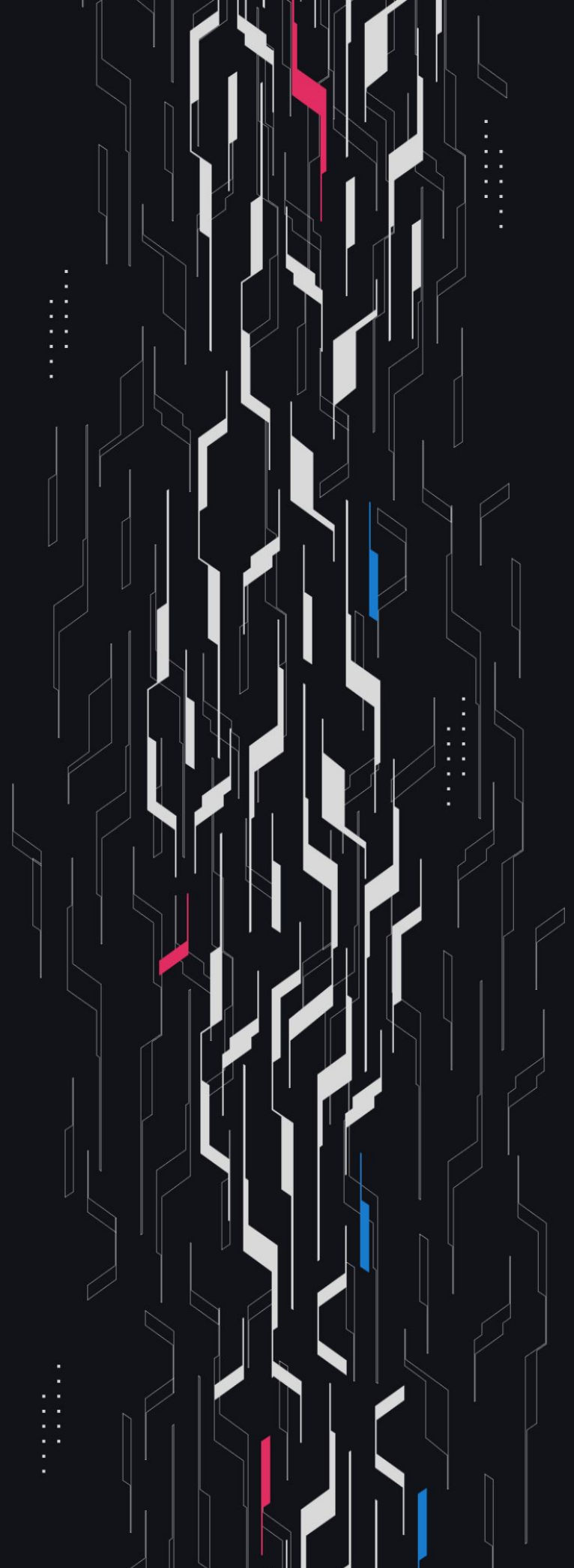
GA GUARDIAN

USDT0

**Mantle Deployment
and Peer Verification**

Security Assessment

November 12th, 2025



Summary

Audit Firm Guardian

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Client Firm USDT0

Final Report Date November 12, 2025

Audit Summary

USDT0 engaged Guardian to review the security of their USDT0's Mantle Deployment and Peer Verification. From the 4th of November to the 5th of November, an auditor reviewed the source code in scope. All findings have been recorded in the following report.

Confidence Ranking

Given the lack of critical issues detected and minimal code changes following the main review, Guardian assigns a Confidence Ranking of 5 to the protocol. Guardian advises the protocol to consider periodic review with future changes. For detailed understanding of the Guardian Confidence Ranking, please see the rubric on the following page.

✓ Verify the authenticity of this report on Guardian's GitHub: <https://github.com/guardianaudits>

Guardian Confidence Ranking

Confidence Ranking	Definition and Recommendation	Risk Profile
5: Very High Confidence	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: Code is highly secure at time of audit. Low risk of latent critical issues.	0 High/Critical findings and few Low/Medium severity findings.
4: High Confidence	Code is clean, well-structured, and adheres to best practices. Only Low or Medium-severity issues were discovered. Design patterns are sound, and test coverage is reasonable. Small changes, such as modifying rounding logic, may introduce new vulnerabilities and should be carefully reviewed. Recommendation: Suitable for deployment after remediations; consider periodic review with changes.	0 High/Critical findings. Varied Low/Medium severity findings.
3: Moderate Confidence	Medium-severity and occasional High-severity issues found. Code is functional, but there are concerning areas (e.g., weak modularity, risky patterns). No critical design flaws, though some patterns could lead to issues in edge cases. Recommendation: Address issues thoroughly and consider a targeted follow-up audit depending on code changes.	1 High finding and ≥ 3 Medium. Varied Low severity findings.
2: Low Confidence	Code shows frequent emergence of Critical/High vulnerabilities (~2/week). Audit revealed recurring anti-patterns, weak test coverage, or unclear logic. These characteristics suggest a high likelihood of latent issues. Recommendation: Post-audit development and a second audit cycle are strongly advised.	2-4 High/Critical findings per engagement week.
1: Very Low Confidence	Code has systemic issues. Multiple High/Critical findings (≥ 5/week), poor security posture, and design flaws that introduce compounding risks. Safety cannot be assured. Recommendation: Halt deployment and seek a comprehensive re-audit after substantial refactoring.	≥ 5 High/Critical findings and overall systemic flaws.

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Project Overview

Project Summary

Project Name	USDT0
Language	Solidity
Codebase	https://github.com/Everdawn-Labs/usdt0-oft-contracts
Commit	Initial commit: 20de3ac6558f4cf9cb438a7bf700a9183279ccc0
Addresses	USDT0-OFT - Implementation: 0xaf37E8B6C9ED7f6318979f56Fc287d76c30847ff USDT0-OFT - OFT: 0xcb768e263FB1C62214E7cab4AA8d036D76dc59CC USDT0-OFT - Proxy Admin: 0x01bFF41798a0BcF287b996046Ca68b395DbC1071 USDT0 - Proxy: 0x779Ded0c9e1022225f8E0630b35a9b54bE713736 USDT0 - Proxy Admin: 0xF555A12bFAEf20cC201A74ae6513Cb4aaDB34B9 USDT0 - Implementation: 0x1EC7df9e74bE05cb5A456ACa2DC1AC2CeC9AB6A3 USDT0 - Safe: 0x4DFF9b5b0143E642a3F63a5bcf2d1C328e600bf8

Audit Summary

Delivery Date	November 12, 2025
Audit Methodology	Static Analysis, Manual Review, Test Suite, Contract Fuzzing

Vulnerability Summary

Vulnerability Level	Total	Pending	Declined	Acknowledged	Partially Resolved	Resolved
● Critical	0	0	0	0	0	0
● High	0	0	0	0	0	0
● Medium	0	0	0	0	0	0
● Low	1	0	0	1	0	0
● Info	1	0	0	1	0	0

Audit Scope & Methodology

Addresses:

USDT0

OFT-contracts:

USDT0 OUpgradable: 0xaf37E8B6C9ED7f6318979f56Fc287d76c30847ff

USDT0 OFT: 0xcb768e263FB1C62214E7cab4AA8d036D76dc59CC

ProxyAdmin: 0x01bFF41798a0BcF287b996046Ca68b395DbC1071

Tether-token:

USDT0 Token: 0x779Ded0c9e1022225f8E0630b35a9b54bE713736

USDT0 Admin: 0xF555A12bfFAEf20cC201A74ae6513Cb4aaDB34B9

USDT0 Impl: 0x1EC7df9e74bE05cb5A456ACa2DC1AC2CeC9AB6A3

Mantle LZ

Endpoint: 0x1a44076050125825900e736c501f859c50fE728c

Send: 0xde19274c009A22921E3966a1Ec868cEba40A5DaC

Receive: 0x8da6512De9379fBF4F09BF520Caf7a85435ed93e

Executor: 0x4Fc3f4A38Acd6E4cC0ccBc04B3Dd1CCAeFd7F3Cd

Chain ID: 5000

Endpoint ID: 30181

Mantle Multisig: 0x4DFF9b5b0143E642a3F63a5bcf2d1C328e600bf8

RPC: <https://rpc.mantle.xyz>

<https://github.com/everdawn-labs/usdt0-tether-contracts-hardhat/pull/87><https://github.com/everdawn-labs/usdt0-oft-contracts/pull/157>

Audit Scope & Methodology

Vulnerability Classifications

Severity	Impact: <i>High</i>	Impact: <i>Medium</i>	Impact: <i>Low</i>
Likelihood: <i>High</i>	● Critical	● High	● Medium
Likelihood: <i>Medium</i>	● High	● Medium	● Low
Likelihood: <i>Low</i>	● Medium	● Low	● Low

Impact

- High** Significant loss of assets in the protocol, significant harm to a group of users, or a core functionality of the protocol is disrupted.
- Medium** A small amount of funds can be lost or ancillary functionality of the protocol is affected. The user or protocol may experience reduced or delayed receipt of intended funds.
- Low** Can lead to any unexpected behavior with some of the protocol's functionalities that is notable but does not meet the criteria for a higher severity.

Likelihood

- High** The attack is possible with reasonable assumptions that mimic on-chain conditions, and the cost of the attack is relatively low compared to the amount gained or the disruption to the protocol.
- Medium** An attack vector that is only possible in uncommon cases or requires a large amount of capital to exercise relative to the amount gained or the disruption to the protocol.
- Low** Unlikely to ever occur in production.

Audit Scope & Methodology

Methodology

Guardian is the ultimate standard for Smart Contract security. An engagement with Guardian entails the following:

- Two competing teams of Guardian security researchers performing an independent review.
- A dedicated fuzzing engineer to construct a comprehensive stateful fuzzing suite for the project.
- An engagement lead security researcher coordinating the 2 teams, performing their own analysis, relaying findings to the client, and orchestrating the testing/verification efforts.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross-referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.
Comprehensive written tests as a part of a code coverage testing suite.
- Contract fuzzing for increased attack resilience.

Findings & Resolutions

ID	Title	Category	Severity	Status
L-01	Mantle Safe Has Different Signers	Warning	Low	Acknowledged
I-01	Mantle Safe Has Different Signers	Warning	Info	Acknowledged

L-01 | Duplicate Nonces On Safe

Category	Severity	Location	Status
Warning	Low		Acknowledged

Description

Mantle Safe API reports duplicate nonce 40, hence there may potentially be a second transaction that will revert with `LZ_SameValue()`. This is due to the Safe API’s faulty pagination, where the count (21) on page 2 exceeds the limit (20) for that page, hence the same nonce appears both in page 2 and 3 of the queued transactions request.

Recommendation

Ensure not to execute a duplicate and notify Safe API of this pagination issue.

Resolution

USDT0 Team: Acknowledged.

I-01 | Mantle Safe Has Different Signers

Category	Severity	Location	Status
Warning	● Info	Global	Acknowledged

Description

The Mantle safe has the following signers:

- 1. 0x488A3d37442F583c1c0a356118d1d6181b22434c
- 2. 0x4192D72c281dB0eF10464dD274EE583C33256ac5
- 3. 0x00F6D2b4B69Ce697f913Da16A9D73283dc4C78F2
- 4. 0x7D8e979dcC6EC933c70FBe28B2B7700907D2aFEA
- 5. 0x5F0DcA105195Bd58277aa4b4CABC90a791E1b982

This differs from Flare and Conflux Safes which use 0x1a6362AD64ccFF5902D46D875B36e8798267d154 instead of 0x00F6D2b4B69Ce697f913Da16A9D73283dc4C78F2

Recommendation

Ensure the signers on the Mantle Safe are intended.

Resolution

USDT0 Team: Acknowledged.

Disclaimer

This report is not, nor should be considered, an “endorsement” or “disapproval” of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any “product” or “asset” created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian’s position is that each company and individual are responsible for their own due diligence and continuous security. Guardian’s goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis. Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract’s safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

About Guardian

Founded in 2022 by DeFi experts, Guardian is a leading audit firm in the DeFi smart contract space. With every audit report, Guardian upholds best-in-class security while achieving our mission to relentlessly secure DeFi.

To learn more, visit <https://guardianaudits.com>

To view our audit portfolio, visit <https://github.com/guardianaudits>

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